

# Action–Evidence Geometry and Execution Certificates

## Supplementary Evidence

### Anonymous ACL submission

This supplement reports the complete benchmark tables, the matched geometry controls, and the reproducibility details for the deterministic DAFR execution layer. The main paper’s claims are limited to typed action–evidence encoding, joint-risk execution constraints, and certificate-based diagnosis and repair. The matched studies do not claim that geometric code is more expressive or intrinsically safer than unrestricted predicates.

#### A Evaluation Protocol

**AgentDojo.** AgentDojo v1.2.2 contains Banking, Slack, Travel, and Workspace. We evaluate DeepSeek V4 Flash, GPT-5.4-mini, and Claude Haiku 4.5 on 71 clean and 585 adversarial cases. The clean/attack counts are 16/144, 21/105, 20/140, and 14/196 for the four suites. Clean utility, attack utility, and ASR are unweighted macro-averages over suites.

**Agent Security Benchmark.** The ASB comparison contains 204 cases with DeepSeek V4 Flash. It reports utility, task success, and ASR. The reported DAFR run uses the benchmark’s source-tagged runtime configuration; the source setting is part of the threat model and should not be generalized to settings in which provenance is unavailable.

**Aggregation and result provenance.** The main AgentDojo protocol uses four suites (71 clean and 585 attacked episodes). The archived three-suite subset contains 57 clean and 389 attacked episodes; its 88.1% clean and 83.2% attack utility for DAFR are different aggregates, not alternative estimates for the four-suite table. For metric  $x$ , suite macro-averaging is  $\bar{x} = \frac{1}{4} \sum_{s=1}^4 x_s$ , rather than pooling all episodes. The four-suite DAFR aggregate

is 87.5% clean and 81.4% attack utility.

The retained GPT baseline table combines earlier three-suite document aggregates with Workspace reruns; the DAFR row is recorded as a four-suite rerun. These baseline rows must not be described as newly reproduced end-to-end runs. The plotted multi-model tables retain archived aggregates; paired statistical comparisons require the underlying episode outcomes. The ASB aggregate specifies seed 17, stratified attack sampling, round-robin case sampling, shared workflow plans, and source-tagged tool provenance. Its task-success score checks completion of required normal tools, while ASR is the stored attack-goal match or an executed attacker-tool trace. These protocol choices limit direct comparison with other ASB settings.

**Execution boundary.** DAFR runs after a planner proposes a concrete call and before the trusted runtime executes it. The deterministic typed field interface validates registered fields and binds them to shared effect roles. The action–evidence encoder retains field identity, evidence source, authorization, and state relations. No language model is used by the execution verifier or by the matched geometry controls.

#### B Matched Geometry and Certificate Studies

The membership arms share the same risk vectors; the repair arms share the same typed actions, evidence, and candidate transformations. Axis-ifelse checks each risk coordinate independently. Geometry evaluates one shared joint-risk region. The equivalent predicate computes the same halfspace and norm inequalities with ordinary arithmetic and is included as an implementation-equivalence con-

trol. A Boolean-only interface has no repair operator. Its zero-repair entry in the archived runner is fixed by construction and is excluded from empirical comparisons here.

The joint-risk suite contains 512 four-dimensional vectors. Every coordinate is below its independent 0.8 threshold, while 25 vectors exceed the shared joint budget. Geometry and the equivalent predicate reject the same 25 vectors. The repair suite contains 256 local write cases: 192 have a safe candidate obtained by removing an unsupported optional destination, and 64 require blocking because the destination is semantically required. Geometry and a predicate supplied with the same repair oracle recover the same 192 actions. The equality is expected because the same executable repair transformation is used in both arms; it does not show an intrinsic geometric repair advantage.

| Backend              | Safe allow | Unsafe block | False allow |
|----------------------|------------|--------------|-------------|
| Axis thresholds      | 487/487    | 0/25         | 25/25       |
| Joint geometry       | 487/487    | 25/25        | 0/25        |
| Equivalent predicate | 487/487    | 25/25        | 0/25        |

Table 1: Joint-risk membership audit: 487 feasible and 25 infeasible vectors.

| Backend + shared repair | Repaired | Required blocks |
|-------------------------|----------|-----------------|
| Geometry                | 192/192  | 64/64           |
| Equivalent predicate    | 192/192  | 64/64           |

Table 2: Local action-repair audit: both backends block all 256 original proposals and preserve the intended effect in all 192 repairable cases (192/256 overall). The same transformation and effect check are used in both arms. A decision-only interface has no repair operator and is not treated as an empirical repair competitor.

We additionally audited 614 saved AgentDojo decision records. Four records were malformed and 610 were well formed. Rechecking the recorded hard margins with an equivalent slack predicate agreed on 610/610 records; 546 were allowed and 64 were blocked or clarified. This is a saved-certificate consistency audit, not a new planner replay, external execution, or utility/ASR measurement.

**Fixed-family stress test.** A 64-case in-process suite crosses fresh and stale state with four fixed injection families: forged approval

or runtime metadata, policy-override instructions, Chinese webpage instructions, and fabricated prerequisite claims. The runtime blocks 64/64 injected writes, including 32/32 fresh-state and 32/32 stale-state cases. An unprotected local executor performs 64/64 writes. The suite has no external side effects and is reported as a mechanism stress test rather than an adaptive-attack result.

## C Complete AgentDojo Results

Values are percentages. Clean and attack utility are higher-is-better; ASR is lower-is-better. The full case counts are given in the protocol above.

| Method       | Clean | Attack | ASR  |
|--------------|-------|--------|------|
| No Defense   | 86.8  | 75.4   | 17.7 |
| Sandwich     | 89.8  | 76.2   | 12.0 |
| Prompt       | 85.0  | 79.2   | 3.9  |
| Spotlighting | 74.4  | 77.8   | 11.3 |
| PIGuard      | 44.9  | 42.5   | 0.0  |
| ProtectAI    | 48.0  | 36.2   | 4.2  |
| Progent      | 75.2  | 71.7   | 2.1  |
| DRIFT        | 62.6  | 53.2   | 1.4  |
| <b>DAFR</b>  | 87.5  | 81.4   | 0.0  |

Table 3: AgentDojo results with DeepSeek V4 Flash.

| Method       | Clean | Attack | ASR |
|--------------|-------|--------|-----|
| No Defense   | 72.2  | 61.2   | 5.2 |
| Sandwich     | 80.6  | 65.4   | 2.1 |
| Prompt       | 76.6  | 65.4   | 0.6 |
| Spotlighting | 70.5  | 62.7   | 2.5 |
| PIGuard      | 46.3  | 34.4   | 0.0 |
| ProtectAI    | 48.9  | 38.6   | 2.3 |
| Progent      | 66.0  | 54.2   | 1.4 |
| DRIFT        | 41.7  | 38.8   | 0.2 |
| <b>DAFR</b>  | 68.7  | 59.5   | 0.0 |

Table 4: AgentDojo results with GPT-5.4-mini.

| Method       | Clean | Attack | ASR |
|--------------|-------|--------|-----|
| No Defense   | 74.1  | 66.3   | 0.2 |
| Sandwich     | 75.4  | 67.1   | 0.0 |
| Prompt       | 70.9  | 65.1   | 0.0 |
| Spotlighting | 74.1  | 62.5   | 0.0 |
| PIGuard      | 51.6  | 38.2   | 0.0 |
| ProtectAI    | 56.0  | 43.9   | 2.1 |
| Progent      | 65.3  | 58.1   | 0.0 |
| DRIFT        | 43.1  | 44.1   | 0.2 |
| <b>DAFR</b>  | 70.4  | 68.4   | 0.0 |

Table 5: AgentDojo results with Claude Haiku 4.5.

Across the three planners, DAFR records 0 observed ASR on the evaluated static attacks. Relative to PIGuard, the other method with 0 observed ASR in all three planner settings, the attack-utility differences are 38.9, 25.1, and

30.2 percentage points. These are descriptive differences on the fixed benchmark cases, not claims of statistical superiority.

## D Agent Security Benchmark

| Method                 | Utility | Task success | ASR        |
|------------------------|---------|--------------|------------|
| No Defense             | 88.7    | 77.9         | 90.2       |
| Delimiter              | 87.3    | 77.0         | 86.3       |
| Instruction Prevention | 88.0    | 77.5         | 39.7       |
| Observation Sandwich   | 88.0    | 77.0         | 27.9       |
| TS-Flow                | 85.5    | 73.5         | 19.1       |
| Progent                | 86.0    | 73.0         | 3.4        |
| DRIFT                  | 85.8    | 73.0         | 23.0       |
| <b>DAFR</b>            | 86.8    | 73.5         | <b>0.0</b> |

Table 6: ASB results on 204 DeepSeek cases.

## E System Ablation and Signed Margins

The system ablation uses the same 71 clean and 585 attacked cases with DeepSeek V4 Flash. These rows remove interacting execution modules and therefore do not isolate the syntax of the geometric boundary. The matched representation study above supplies that narrower comparison.

| Configuration    | Clean | Attack | ASR  |
|------------------|-------|--------|------|
| DAFR             | 87.5  | 81.4   | 0.0  |
| w/o projection   | 79.2  | 81.8   | 0.0  |
| w/o encoding     | 91.5  | 85.3   | 0.7  |
| w/o defense path | 90.2  | 81.3   | 16.7 |
| w/o feedback     | 79.4  | 80.0   | 0.8  |

Table 7: AgentDojo system ablation.

| Setting              | Calls | Admit | Reject | Gap   |
|----------------------|-------|-------|--------|-------|
| AgentDojo / DeepSeek | 3947  | 0.712 | -0.368 | 1.080 |
| AgentDojo / GPT      | 2916  | 0.748 | -0.214 | 0.963 |
| AgentDojo / Claude   | 2220  | 0.750 | -0.289 | 1.039 |
| ASB / DeepSeek       | 488   | 0.134 | -0.885 | 1.018 |

Table 8: Signed margins;  $M_t$  is the minimum hard-constraint slack.

The admitted/rejected sign separation is a consistency check for the decision rule. It is not a calibration or security theorem. The local matched suite and saved-certificate audit are therefore reported separately from benchmark utility and ASR.

## F Reproducibility Record

The evaluated configuration uses seven shared effect roles, 45 action-evidence coordinates, 13

hard constraint templates, and one soft confidence facet. Coefficients and risk budgets are configuration constants; freezing them before scoring does not substitute for a parameter-sensitivity analysis. Every decision records the concrete call, selected constraints, margin vector, source links, and final decision. The benchmark comparisons use fixed case sets and report malformed or blocked calls through the benchmark’s native scoring functions. No matched geometry case invokes a real external tool.

The repository contains the benchmark manifests, raw decision records, aggregation scripts, and the matched geometry runner. The local suites should be interpreted as mechanism tests: they establish coupled-risk and certificate behavior under controlled inputs, while AgentDojo and ASB provide the task-level utility and observed attack outcomes.